

Generated locally on 2026-09-12 from docs/LOCAL-INTEGRATION.md

AHP Local Integration Matrix

Last updated: 2026-09-12.

Local dev database: existing Docker PostgreSQL `ahealth-pg`, database `ahealth_dev`.

Isolated test database: `ahealth_test` on the same `ahealth-pg` container.

Local apps: doctor-admin `3100`, admin `3200`, analytics `3300`.

Private pre-write backup taken before dev DB mutations:

`~/ahp-local-backups/ahealth_dev_20260912_133957.dump`.

Checkpoint

- Completed:
- PASS: BFF/server auth refresh for doctor-admin, admin and analytics.
 - PASS: Same-browser cookie isolation for all three apps.
 - PASS: CBC evidence contradiction resolved against exact DB/schema and API.
 - PASS: Core synthetic doctor workflow through API persistence/read-back.
 - PASS: Admin management actions where fixtures exist.
 - PASS: Analytics API reads for current synthetic data.
 - PASS: Isolated `ahealth_test` migrated and auth tests run against it.
 - PASS: Firefox browser automation installed and used for real login checks.
 - PASS: Safe local lifecycle scripts fixed and verified.
 - PASS: Production builds for doctor-admin, admin and analytics.
 - PASS: Payment refund API verified against `ahealth_test`.
- Remaining:
- MOCKED: external AI, SMS/notification delivery, payment providers and gateway-provider integrations stay local/mock.

Auth Lifecycle

Case	Apps	Status	Evidence
Valid session/access token	doctor-admin, admin, analytics	PASS	Firefox visited <code>/queue</code> , <code>/emergency</code> , <code>/surveillance</code> in one browser context; <code>/bff/users/me</code> returned roles <code>clinician</code> , <code>platform_admin</code> , <code>platform_admin</code> .
Expired access + valid refresh	doctor-admin, admin, analytics	PASS	Synthetic expired NextAuth cookies returned BFF <code>200</code> , emitted a rotated <code>Set-Cookie</code> , and the next BFF call with that replacement cookie returned <code>200</code> .
Invalid/revoked refresh	doctor-admin, admin, analytics	PASS	Expired cookie with invalid refresh returned <code>401</code> and English message <code>Your session has expired. Please sign in again.</code>

Case	Apps	Status	Evidence
Concurrent refresh	doctor-admin	PASS	Two simultaneous <code>/bff/users/me</code> calls with the same expired doctor cookie both returned <code>200</code> , both set replacement cookies, role stayed <code>clinician</code> .
Three apps in one browser context	doctor-admin, admin, analytics	PASS	Firefox cookie jar held <code>a-health-admin.session-token</code> , <code>a-health-analytics.session-token</code> , <code>a-health-doctor.session-token</code> ; no role confusion on BFF reads.
Password preservation	login forms	PASS	Doctor login no longer trims password before <code>signIn</code> ; admin/analytics already preserved password value.
Real Firefox login forms	doctor-admin, admin, analytics	PASS	Playwright Firefox used real login forms with no injected cookies. Callback responses were <code>200</code> with protected callback URLs <code>/queue</code> , <code>/emergency</code> , <code>/surveillance</code> ; refresh preserved roles <code>clinician</code> , <code>platform_admin</code> , <code>platform_admin</code> in one browser context.

CBC Evidence

Check	Status	Evidence
Running diagnostics service DB/schema	PASS	Diagnostics process cwd is <code>services/diagnostics</code> ; service <code>.env</code> points to <code>ahealth_dev</code> ; exact query used <code>ahealth_dev.public</code> .
Exact DB result	PASS	<code>ahealth_dev.public.investigation_orders</code> contains ID <code>34adf903-50e5-4ad6-bd91-9cedfb08b1f5</code> ; code <code>Complete Blood Count (CBC)</code> , type <code>laboratory</code> , status <code>ordered</code> , linked thread/patient present.
Exact API result	PASS	<code>GET /investigation-orders/34adf903-50e5-4ad6-bd91-9cedfb08b1f5</code> on diagnostics returned the same ID/code/type/status.
Contradictory no-row query	RESOLVED	<code>ahealth_test.public</code> has zero rows for that ID; the earlier no-row result was from a different DB context, not evidence that the dev order was gone.

Doctor Workflows

Workflow	Status	Evidence
Consultation -> messaging -> patient record	PASS	Synthetic consultations created/offered/accepted; 3 completed synthetic consultations persisted. Synthetic messages count: 3; read-back by care thread succeeded during API run.
Diagnostic ordering/list/detail	PASS	3 synthetic diagnostic orders persisted with <code>clinical_notes='Synthetic local integration order only'</code> ; order detail read-back returned <code>ordered</code> . Admin attempt to create an order returned <code>403</code> .
Prescription creation -> persistence -> patient record	PASS	3 prescriptions persisted with item <code>Synthetic ORS template</code> ; API read-back by prescription ID and patient prescription list confirmed the created prescription.
Follow-up creation/read/check-ins	PASS	3 follow-up cycles persisted from completed synthetic consultations; cycle read-back returned <code>active</code> ; adherence rows generated from prescriptions. Doctor patient-specific adherence filter returned <code>403</code> , preserving role boundary.
Availability/slots -> authorized appointment workflow	PASS	Doctor availability set true; slot publish/read-back succeeded for slot <code>b7a09df7-16e0-45b7-9f55-8e281e401ab8</code> ; appointment <code>01d89e80-b4ac-4d7e-a319-bf1fba0a2b1e</code> booked/read back as booked; early start returned <code>409</code> .
Consultation completion restrictions	PASS	Repeating <code>complete</code> on a completed synthetic consultation returned <code>409</code> .
Lab-result boundary	PASS	Doctor can order/read own orders; result filing remains diagnostics/lab workflow and was not bypassed from doctor-admin.

Admin Workflows

Workflow	Status	Evidence
Clinician verification	PASS	Pending clinician was approved; verified clinician count is now 2.
Device management	PASS	Admin vehicle sensor registration/revocation completed; DB aggregate shows 1 synthetic <code>SYN-%</code> device with status <code>revoked</code> . Patient-worn device attempt without patient profile returned <code>403</code> .
Emergency dispatch/status	PASS	Synthetic emergency <code>b75b84bc-3c1c-4f43-8c8a-8be451cb73cb</code> advanced <code>dispatched -> en_route -> arrived -> resolved</code> ; read-back returned <code>resolved</code> . Direct <code>dispatched -> resolved</code> returned <code>409</code> .

Workflow	Status	Evidence
Payment refund	PASS	Isolated payment service ran on port 5012 with DATABASE_URL=...ahealth_test ; API created a synthetic cash intent (201 , succeeded), refunded 2345 (201), DB read-back showed partially_refunded and refunded_amount=2345 , excess refund returned 422 VALIDATION_FAILED ; fixture rows were cleaned.

Analytics

Workflow	Status	Evidence
Surveillance filters/aggregates	PASS	/surveillance/conditions and /surveillance/trends returned successfully under analytics/admin token during API run; Firefox rendered protected surveillance page.
Research datasets	PASS	/research/datasets returned successfully under analytics/admin token during API run. External research export/provider integrations remain mocked/local.

Service Accounting

Service	Status	Notes
auth	PASS	Login, refresh, invalid refresh, concurrent refresh and tests verified.
patient	PASS	Patient profile ownership used by consultation, appointment, prescription and follow-up workflows.
doctor	PASS	Availability, slots, clinician verification and clinician reads verified.
appointment	PASS	Book/read/early-start restriction verified.
consultation	PASS	Create/queue/accept/complete/prescription/follow-up state restrictions verified.
messaging	PASS	Persisted care-thread message and read-back verified.
followup	PASS	Cycle/adherence creation and role-boundary rejection verified.
diagnostics	PASS	Order create/detail and CBC exact-ID evidence verified.
admin-facing devices/emergency/facilities	PASS	Device and emergency actions verified; facility used as dispatch destination.
payment	PASS	Refund API and service tests verified against ahealth_test ; provider remains local/mock.

Service	Status	Notes
insurance	IMPLEMENTED BUT UNVERIFIED	Service is routed but no end-to-end claim fixture was exercised.
notification	MOCKED	Local worker/templates present; real SMS/app push delivery not used.
sync	IMPLEMENTED BUT UNVERIFIED	Service is routed; offline sync batch not exercised in this pass.
gateway	MOCKED	Gateway routes exist with signatures; real USSD/SMS providers not used.
ai	MOCKED	Local AI service only; no external AI provider used.
research/surveillance	PASS	Analytics reads verified against current synthetic data.

Automated Verification

Check	Status	Result
ahealth_test migration	PASS	Prisma applied all 12 migrations to ahealth_test .
Auth tests on isolated DB	PASS	DATABASE_URL=...ahealth_test pnpm --filter @a-health/auth test : 15 pass, 0 fail.
Payment tests on isolated DB	PASS	DATABASE_URL=...ahealth_test pnpm --filter @a-health/payment test : 15 pass, 0 fail. Cleanup now removes failed-intent fixtures for synthetic users.
App type-checks	PASS	pnpm --filter doctor-dashboard type-check, admin-console type-check, analytics-portal type-check all passed after login/build fixes.
Auth build	PASS	pnpm --filter @a-health/auth build passed.
Browser automation	PASS	Firefox 155 installed under .playwright-browsers ; real login forms verified for doctor/admin/analytics in one context, no cookie injection.
Production builds	PASS	Actual webpack diagnostic was build-time next/font/google fetch failure for IBM Plex fonts under restricted network. Replaced with local system font CSS variables and disabled webpack build worker for visible diagnostics. pnpm --filter doctor-dashboard build, admin-console build, analytics-portal build all passed.

Safe Local Lifecycle

Use these scripts from the repo root:

```
scripts/local/start.sh all
scripts/local/status.sh
scripts/local/stop.sh
```

Safety behavior:

- Reuses existing `ahealth-pg` ; does not start the Compose PostgreSQL service.
- Uses gateway port `4021` , matching service env and Compose.
- Records AHP-owned tmux sessions, visible PIDs, or namespace-safe `port:<port>` listeners under `.dev-logs/pids` .
- Removes stale PID files and converts invisible-but-listening services to port-only records instead of reporting false stale failures.
- Refuses duplicate starts when a port is held by a non-AHP process.
- Stops only AHP-owned tmux sessions or processes whose `cwd/cmdline` identify them as AHP-owned.
- Leaves already-running listeners alone when the listener PID is not visible in the current process namespace.
- Does not use `fuser -k` , volume deletion, database reset, or destructive migrations.

Verification on 2026-09-12:

Check	Status	Evidence
PID/status mismatch	PASS	Root cause was mixed process visibility: the running AHP stack was listening on ports but several listener PIDs were not visible from the managed execution namespace, so old numeric PID files were stale even though services were alive. <code>status.sh</code> now reports these as running port-only records.
<code>stop -> status -> start -> status</code>	PASS	<code>stop.sh</code> preserved existing port-only listeners, stopped only AHP-owned tmux sessions, removed stale records, and <code>start.sh</code> all restarted missing services without duplicate launches.
Backend health	PASS	<code>/health</code> returned HTTP <code>200</code> for every service port <code>4001</code> through <code>4025</code> , including patient <code>4002</code> , notification <code>4008</code> , insurance <code>4018</code> , gateway <code>4021</code> , and sync <code>4022</code> .
App availability	PASS	Apps on <code>3100</code> , <code>3200</code> , and <code>3300</code> responded with protected-route redirects (<code>307</code>), matching authenticated-app routing expectations.
Container preservation	PASS	<code>docker ps</code> showed <code>ahealth-pg</code> , <code>dvwa-db-1</code> , and <code>dvwa-dvwa-1</code> still running; no Compose PostgreSQL was started and no DVWA container was stopped.
Sandbox detach behavior	BLOCKED	Detached processes and tmux sessions created from a finished managed command are reaped by the executor environment. For this acceptance-prep session, missing services were held in a live foreground supervisor while health was verified; running the same scripts from a normal local shell should keep tmux sessions alive.